Unit 1: Supervised Learning Fundamentals
- Linear regression, ordinary least squares, and cost function optimization
- Logistic regression for binary classification and decision boundaries
- Regularization techniques: Ridge (L2) and Lasso (L1) regression

Unit 2: Decision Trees and Ensembles
- Decision trees: splitting criteria, entropy, and information gain
- Random forests: bootstrap aggregating and out-of-bag error
- Gradient boosting machines (GBM) and AdaBoost

Unit 3: Support Vector Machines & Kernels
- Linear support vector classifiers and margin maximization
- Kernel trick: polynomial, radial basis function (RBF), and sigmoid kernels
- Soft margin optimization and slack variables

Unit 4: Unsupervised Learning
- K-means clustering: initialization, convergence, and elbow method
- Hierarchical clustering: agglomerative vs divisive dendrograms
- Principal Component Analysis (PCA) for dimensionality reduction

Unit 5: Model Evaluation and Validation
- Cross-validation: k-fold, stratified k-fold, and leave-one-out
- Evaluation metrics: accuracy, precision, recall, F1-score, and ROC-AUC
- Hyperparameter tuning using grid search and random search